

Program Question_6

```
Implicit none
Logical::p,q,r
Logical::exp1,exp2
Integer::i
Logical,Dimension(8)::all_true1,all_true2
Write(*,*)" 'Truth Table for (p ^ q) v r and (p v r) ^ (q v r)"
Write(*,*)" p   q   r | ((p v q) v r)   (p v r)^(q v r)"
Do i=0,7
    !Assign truth values based on binary rep. of i
    p=btest(i,2)
    q=btest(i,1)
    r=btest(i,0)

    !Evaluate the Expression
    exp1=(p.and.q).or.r
    exp2=(p.or.q).And.(q .or.r)
    Write(*,'(L3, L4, L4, A, L10, L10)')p,q,r, "|", exp1,exp2
    !Check for tautology
    all_true1(i+1)=exp1
    all_true2(i+1)=exp2

End Do
Write(*,*)" "
!Check if all values are true
if(all(all_true1))then
    Write(*,*)" (p ^ q) v r is tautology"
else
    Write(*,*)" (p ^ q) v r is not  tautology"

end if
Write(*,*)" "

if(all(all_true2))then
    Write(*,*)" (p v r)^(q v r) is tautology"
else
    Write(*,*)" (p v r)^(q v r) is not  tautology"

end if

End Program
```